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Crosscutting Areas

Dynamic Pricing Under Self-Exciting Arrival Processes

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Abstract. We study dynamic pricing under a stochastic, nonlinear, self-exciting demand arrival process over a finite sales horizon. We adopt such a correlated demand process to capture the phenomenon that customers who have made a purchase can inform and excite future customers to arrive. Specifically, the stochastic arrival intensity is boosted immediately after any purchase and gradually decays between consecutive purchases. The arrival intensity is also affected by the market phase the seller operates in, a growth or saturation stage, which convexly or concavely shapes the number of new customers that can be stimulated by earlier buyers. We show that the optimal pricing policy depends on the current time and excitement level, which measures the currently accumulated influence that earlier buyers exert and is shown to be the only sufficient statistic for our problem. Over time, the seller tends to initially set a low price to attract early buyers, who can excite later arrivals, and later increase the price to maximize profits. Perhaps surprisingly, at a given time, the optimal price rises with the excitement level if the market is in a growth stage, whereas it drops with the excitement level if the market is in a saturation stage. Moreover, we show that adopting the optimal time-varying (open-loop) pricing policy obtained from the deterministic problem (without re-solving the problem) incurs only a constant performance loss relative to the optimal dynamic (i.e., contingent) pricing policy, which would not hold for the classic revenue management problem with a capacity constraint. Finally, we extend our analysis to account for discrete prices and multiple products in an assortment and make a connection between the self-exciting demand arrival process and the immigration-birth process.

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1. Introduction

In recent years, firms have increasingly leveraged customer-generated content to enhance their sales and pricing policies. For instance, customers' enthusiastic sharing of their purchases has become a crucial driver of sales. The organic reach and authenticity of customer-shared content can enhance the brand's visibility and credibility, enticing new customers and driving sales. By monitoring social media platforms, firms can identify trends and adjust pricing policies in real time to maximize profits. This reactive approach has proven effective, because customers create and share a significant portion of product-related content on platforms like Instagram, Facebook, and TikTok.

However, customer-shared content on social media platforms can lose relevance and engagement over time as new content emerges and older posts become less visible. The duration of user engagement, measured through metrics such as likes, comments, saves, and views, varies across different platforms. For instance, on Snapchat, engagement drops off almost immediately, exhibiting what can be termed "Instant Decay." On Twitter, the engagement span is approximately 43 minutes. Facebook content tends to engage users for about 76 minutes, whereas Instagram posts remain relevant for up to 19.75 hours. LinkedIn posts have a longer lifespan, with engagement persisting for about 24 hours (Graffius 2024).

The rapid growth of social media, with the network effects of word-of-mouth marketing and the associated decay, has significantly shaped how customers influence potential buyers. Customers who have purchased

a product share their experiences within their social networks. This sharing leverages the social proof provided by satisfied customers to attract new buyers. However, as mentioned, this shared content can quickly decay and lose its impact. Companies need to respond actively to such a dynamic process of customer influence.

We observe how customers influence potential buyers using a real data set. For example, we obtain a good model fit when calibrating the arrival process of a sunscreen product on JD.com, one of China's largest retailers, for March 8 by adopting a self-exciting process, also known as the Hawkes process. Each customer purchase stimulates additional purchases. Such a stimulation can gradually decay over time, as shown in Figure 1(a). In addition, we simulate a sample path of the customer-generating process for this product to illustrate the transition from one "generation" of purchased consumers to the next one being influenced, as shown in Figure 1(b). In this figure, cross signs denote the arrival of customers who made a purchase, whereas arrows indicate transitions between generations.

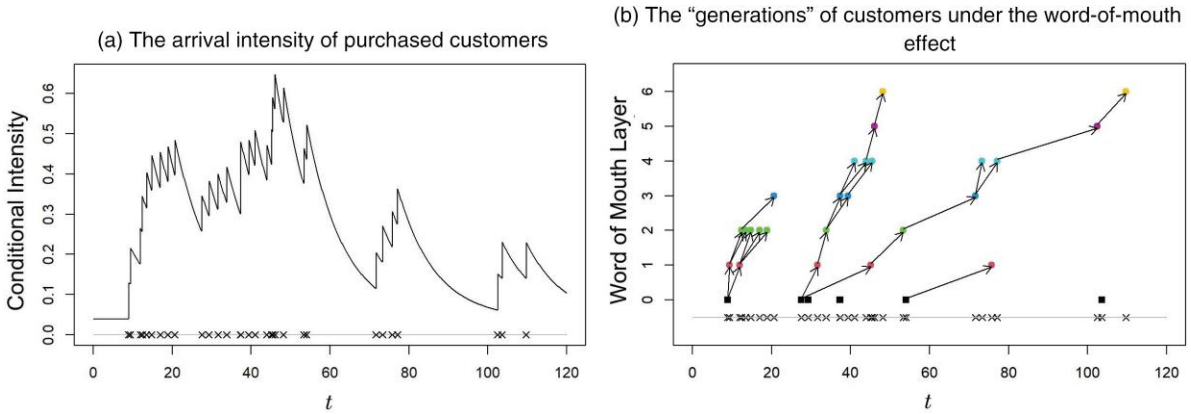
When customer arrivals are self-exciting, the firm may want to adjust the product price as the dynamic purchase process evolves. It is challenging to balance the current period's profit with future potential profit in such a dynamic setting. Lower prices can attract more buyers and foster future sales, whereas higher prices can improve immediate profitability. It is essential to strike a nuanced balance between immediate payoffs and future benefits.

In this paper, we study how a seller should dynamically adjust the selling price under a generalized self-exciting customer arrival process, in which earlier buyers excite potential future customers. Each new purchase continuously stimulates additional purchases, thereby increasing the stochastic arrival intensities, a feature we refer to as self-excitement of the arrival process. Moreover, such stimulation may decay over time because the influence of earlier purchases tends to diminish as the associated social media posts become less fresh. Upon arrival, a customer decides whether to make a purchase based on the current price. We formulate this problem as a dynamic program and study the structural properties of the optimal pricing policy. We show that the optimal pricing policy depends on the current time and excitement level, which measures the currently accumulated influence that earlier buyers exert and is shown to be the only sufficient statistic for our dynamic pricing problem. We fully characterize how the optimal pricing policy depends on the current time and excitement level.

We first demonstrate that the optimal pricing policy consistently encourages a higher purchase rate as the excitement level increases. This finding says that at any time, if the influence that earlier buyers exert becomes higher, the optimal purchase rate should also be higher; although the optimal pricing can either increase or decrease in the excitement level, the optimal purchase rate always increases as the excitement level rises, even if the seller decides to increase the price.

Next, we show that the optimal price may not be monotone in the excitement level as the state variable. This nonmonotonic behavior differs fundamentally from results in the existing revenue management (RM) literature, where the optimal price is typically constant without capacity constraints or is monotone in the capacity level that is the state variable. As excitement levels increase, it might be expected that the seller would raise the price to maximize profit. However, we find that the optimal price increases (respectively, decreases) with the excitement level, depending on whether the market is in a growth (respectively, saturation) phase. During a rapid growth phase of a product, the optimal price decreases with the excitement level because a small increase in excitement significantly expands the customer base, thereby convexly increasing potential future arrivals. Therefore, when the excitement level is higher, the seller should reduce the price further to build an even larger pool of earlier buyers. Conversely, during the

Figure 1. (Color online) Sales Process on JD.com



saturation phase of a product, the optimal price increases with the excitement level, because the limited pool of potential customers due to market saturation diminishes the benefit of lowering prices to entice future arrivals. In this case, when the excitement level is higher, the seller increases the price to generate higher immediate profit and exploit the larger pool of earlier buyers who can excite future customers. For a linear self-exciting function, we show that the optimal intensity is independent of previous purchases and can be obtained by solving a set of two ordinary differential equations (ODEs).

Third, we demonstrate that the optimal price process is a submartingale; that is, it exhibits an upward trend, in the absence of excitement decay. At the beginning of the sales horizon, the seller tends to set a low price to attract purchases, thereby generating additional potential arrivals in the future. As the end of the sales horizon approaches, the seller tends to raise the price to make immediate profits. As prices are expected to rise stochastically over time, such an upward trend aligns with offering early-bird discounts each morning to quickly attract a large customer base, anticipating an increase in social media-driven arrivals as the day progresses.

Computing the optimal pricing policy in our setting can be challenging. Unlike traditional RM problems, where inventory is finite and decreases over time, the excitation level in our problem can be uncapped. This makes numerical computation challenging, as a large number of scenarios must be accounted for. This would be especially difficult for problems in which the base arrival rate is large. To search for easy-to-compute heuristics and study their performance, we consider a series of problems in which we scale up the arrival rate by a factor of n. We show that the expectation of the excitement level grows at a rate of n if the self-exciting function is Lipschitz. Under the latter innocuous assumption, the variance of the optimal intensity vanishes to zero at a rate of $1/n$. Thus, as the scale of the problem increases, the optimal intensity converges to its expected value in probability. Because the expectation of the optimal intensity is, in general, intractable, we are motivated to seek deterministic heuristics in which the intensity over time is predetermined and does not change contingently on the excitement level.

In traditional RM problems, a commonly used deterministic heuristic is derived from the corresponding deterministic problem where customers arrive in a fluid fashion. We take a similar approach and study the deterministic version of our problem. For the traditional inventory capacitated RM problem, Wang and Wang (2022) show that the difference in revenues between the stochastic and deterministic problems is at the magnitude of $\ln(n)$. However, we prove that the difference in our setup is bounded by a constant independent of n. We then apply the deterministic heuristic, which is a time-varying (open-loop) pricing policy, to the stochastic problem and show that its performance loss is also a constant independent of n, whereas the performance loss of such a heuristic is at the scale of $\sqrt{n}$ for the traditional RM problem (Gallego and Van Ryzin 1994).

We then extend our base model to three extensions: discrete prices, immigration-birth process, and multiple products. First, we establish that the synthetic discrete price heuristic is not asymptotically optimal, in contrast to the classic RM literature. Specifically, any set of K discrete prices fails to synthesize and align with the optimal price, even if their expected price matches the optimal price. Therefore, pricing managers are better off constantly monitoring the market and adjusting prices without being constrained by predetermined price sets. Second, we show that the self-exciting process with a linear self-exciting function is equivalent to an immigration-birth process. Unlike the traditional immigration-birth literature, in which the descendant population may grow explosively, we show that the optimal pricing decision can ensure that each "immigrant" produces a manageable number of descendants. Specifically, by dynamically adjusting prices, managers effectively limit the number of descendants each immigrant generates, thereby profitably maintaining control over the sales process and preventing low-margin runaway growth. Third, for customers making a selection among multiple products, we show that most of our findings from the base model remain applicable to the multiproduct pricing setting, in which a purchase of one product can attract future arrivals who may end up purchasing another product.

Finally, we conduct an extensive numerical study to derive further insights. First, we investigate the necessity of considering self-excitement when the seller makes pricing decisions. Our findings indicate that neglecting self-excitation leads to significant profit loss for the seller. This insight underscores the need for pricing managers to consider the word-of-mouth effect in their pricing policies. Second, we quantify the value of dynamic (more precisely, contingent) pricing by comparing optimal dynamic pricing against optimal static pricing. Our results demonstrate that implementing optimal dynamic pricing can significantly enhance the seller's profit by adjusting prices in response to information dissemination dynamics across social networks. Third, we study the performance of our theoretically examined deterministic heuristic, obtained from solving the deterministic counterpart of the original stochastic problem, and observe its superiority in achieving profitability.

2. Literature Review

This paper is closely related to four distinct streams of literature: self-exciting processes, correlated customer arrivals in RM, pricing under network effects, and asymptotic analysis in RM. We will briefly review each of these areas and then summarize our contributions. First, our paper is related to the literature on self-exciting processes, also known as Hawkes processes, which were first introduced by Hawkes (1971). Hawkes introduces a family of models for stochastic point processes known as self-exciting and mutually exciting point processes. The process exhibits the critical property that the occurrence of any event increases the probability of the subsequent events. Brémaud and Massoulié (1996) extend this to a nonlinear self-exciting (Hawkes) process, proving the existence of a unique stationary version and the convergence of nonstationary processes. None of these studies has linked the self-exciting process with customer arrivals driven by a seller's pricing decisions. To our knowledge, ours is the first study to model customer arrivals as a self-exciting process in a pricing setting.

Second, our paper is related to the literature on correlated customer arrivals in RM. Typically, customers are assumed to arrive following a Poisson arrival process (Gallego and Van Ryzin 1994). In our setting, arrivals are also influenced by previous purchases. Other studies have modeled customer arrivals using a Markov process. Rustichini and Wolinsky (1995) investigate pricing decisions under a Markov demand process with constant transition probabilities. Aviv and Pazgal (2005) examine dynamic pricing with a partially observed Markovian process, allowing the seller to update probabilities periodically. Keskin and Li (2024) study a firm's dynamic pricing in the presence of unknown, time-varying customer preferences for quality. Jiang (2023) considers a network RM decision and assumes customer arrivals over different periods follow a time-inhomogeneous Markov chain. Our approach differs in considering a continuous-time setting rather than making decisions over discrete time epochs. Specifically, we introduce an excitement level to capture the continuous influence of earlier buyers on new arrivals, which can be viewed as the state in a continuous-time Markov process. Additionally, we demonstrate the monotonicity properties of the optimal pricing policies and show that the price process tends to increase over time.

More recently, another stream of work aims to address the large demand volume and the large variability. Bai et al. (2023) consider a demand model in which customers arrive according to a Markov process, thereby allowing the total demand to follow an arbitrary distribution. The authors propose a fluid approximation of the RM problem that becomes asymptotically tight as resource capacities increase. Li et al. (2025) study an RM model in which customers arrive in multiple stages in the selling horizon. They present asymptotically optimal policies based on a fluid approximation as the number of stages and resource capacities scale up. In contrast to those two papers, which assume discrete time and capacity constraints, our work considers a self-exciting arrival process in continuous time with no capacity constraint. Moreover, in our setting, the optimal policy needs to know the timing of all previous arrivals due to the influence-decay effect of prior purchases.

Third, our paper contributes to the literature on dynamic pricing under network effects. Xie and Sirbu (1995) examine dynamic pricing and its profitability for incumbents and entrants under network externalities and competition, considering the general network benefits dependent on all customers' behaviors. Ajorlou et al. (2018) study a monopolist's dynamic pricing strategy with word-of-mouth information spread over an infinite horizon in a deterministic setting, demonstrating that optimal prices drop to zero infinitely often with negligible marginal costs. Unlike these static or dynamic but deterministic models, our research examines contingent dynamic pricing under demand uncertainty. Another related stream of research focuses on network effects under uncertainty. Lin et al. (2024) integrate diffusion effects for online content into the candidate generation and promotion optimization problem, demonstrating the consistency and efficiency of D-OLS estimators. Jiang et al. (2022) examine the dynamic competition between promotional efforts and pricing by retailers, finding that equilibrium prices increase over time, whereas promotional efforts decrease, highlighting the need for retailers to balance this trade-off intertemporally. Yang and Zhang (2022) explore dynamic pricing and inventory management where customer reviews influence future purchasing behaviors and characterize the optimal policy. Huang et al. (2022) analyze the value of price discrimination based on customers' positions in social networks, showing significant value in small networks but little value in large random networks. Ahn et al. (2025) study how a company can use an influencer to influence its followers to make purchases, finding that the noisy signal of product quality can benefit the company. Our study differs by adopting the self-exciting process to capture how past purchases excite future customer arrivals.

In addition, our paper is related to the literature on dynamic pricing under Bass diffusion models. Dynamic pricing in the context of Bass-type adoption dynamics has gained traction in operations management (Shen et al. 2014, Li 2020). These models assume deterministic adoptions and do not capture the randomness of customer arrivals. More recently, Zhang et al. (2022) model new product adoption as a continuous-time Markov

chain. They propose pricing algorithms that update parameter estimates over time and demonstrate that these algorithms incur a revenue loss of order logarithmic in the market size. In their model, the rate of adoption depends on a limited market size and the proportion of customers who have adopted. In contrast, the Hawkes process in our model allows for an unbounded market size. Unlike models constrained by a finite market size, in which adoption rates decline as the proportion of adopters grows, the Hawkes process itself incorporates the decay of adoption rates over time. Furthermore, each prior purchase stimulates future adoptions in a general way by bringing in additional potential customers.

Finally, in search of heuristics and to analyze their asymptotic performances, we draw inspiration from relevant papers in traditional RM, with which our problem shares similarities. Gallego and Van Ryzin (1994) adopt the optimal solution of the deterministic fluid model as the deterministic heuristic for the stochastic problem. They show that the deterministic heuristic is asymptotically optimal as the demand rate and the inventory are scaled up. As the scale of the problem n grows, the deterministic heuristic has a revenue loss at the magnitude of $\sqrt{n}$. To further reduce the gap, a commonly used heuristic is to periodically resolve the deterministic problem whose solution is then adopted in the stochastic problem (Cooper 2002, Maglaras and Meissner 2006, Reiman and Wang 2008, Jasin and Kumar 2013). In particular, for the price-based RM problem, Jasin (2014) shows that the frequent re-solving heuristic reduces the revenue gap to the magnitude of $\ln(n)$. They also propose a linear rate correction heuristic that adjusts prices according to a predetermined formula and show that it incurs the same $\ln(n)$ performance loss as the re-solving heuristic. Chen et al. (2016) show that a heuristic that adjusts prices of only a small set of products can also achieve the same bound. Studying the dynamic pricing problem under a cascade click demand model where customers are presented with products sequentially, Najafi et al. (2024) demonstrate that the deterministic heuristic is asymptotically optimal and has a performance loss at the magnitude of $\sqrt{n}$. They propose a cascade correction heuristic that updates prices periodically and show that it incurs an expected performance loss of magnitude $\ln(n)$.

To derive asymptotic performance bounds, we bound the variance of the general Hawkes process as the purchase rate scales up. Many previous papers study the limiting behavior of Hawkes processes, most of which consider the setting where time goes to infinity. Bacry et al. (2013) prove the law of large numbers and the central limit theorem for the linear Hawkes process when the time is large. Zhu (2013) obtains the central limit theorem for the nonlinear Hawkes process. The work closest to ours is Gao and Zhu (2018), who prove a central limit theorem for a linear Hawkes process under a scaling regime for the baseline intensity. However, our analysis departs from theirs in two ways: first, we consider a general nonlinear Hawkes process, which introduces greater analytical challenges; second, the process rate in our model is endogenously controlled through a dynamic purchase probability. We show that, despite these complexities, the variance remains at the scale of n as long as the variance of the purchase probability is not too large.

3. Formulation and Preliminary Results

3.1. Model

Consider a seller who dynamically adjusts prices to promote and sell a single product over a finite horizon, denoted as [0, T], say 24 hours. There are two types of potential customers. One type, called organic customers, arrives randomly according to their own needs with an arrival rate φ. The other type of customers is excited by previous customers who have made a purchase and have a strong incentive to inform others, for example, by posting about their purchase on social media. Let $\{t_1, t_2, \ldots, t_k\}$ represent the realized and observed sequence of arrival times of purchasing customers up to time t. For a customer who arrives and makes a purchase at time t_i, this earlier buyer influences additional customers, referred to as influenced customers, who may arrive at a later time $t \geq t_i$, at a stochastic rate of $h(\sum_{t_i<t} \alpha e^{-\beta(t-t_i)})$, where the excitation intensity α represents the baseline contribution to the arrival intensity. As older posts become less visible, the decay parameter β captures the rate at which the excitation decays over time.[1] The function $h(\cdot)$ is strictly increasing, as a rise in the number of those baseline arrivals $\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$ results in a higher arrival rate of influenced customers. Later in this section, we will demonstrate that the optimal pricing strategies exhibit distinct structural properties depending on the shape of $h(\cdot)$.[2]

We now define the stochastic intensity of potential customers, which follows a generalized self-exciting process (Brémaud and Massoulié 1996, Zhu 2015), as follows.

Definition 1 (General Self-Exciting Process). Let $t_1, t_2, \ldots, t_k \in [0, t)$ be the purchase time epochs up to time t. Potential customers arrive according to a point process N_t with the following stochastic intensity:

$$\Phi_t(\mathcal{H}_t) = \underbrace{\varphi}_{\text{organic}} + \underbrace{h\left(\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}\right)}_{\text{influenced}},$$

where the self-exciting function $h(\cdot)$ is strictly increasing.

The above definition of the potential demand process covers a few notable special cases. For instance, when $\alpha = 0$, the demand process reduces to a standard Poisson process. When β goes to infinity, the incentive to inform others quickly decays, and the demand process also reduces to a standard Poisson process. When $h(\cdot) = 0$, there is no influence made by the earlier buyers, and the demand process is again reduced to a standard Poisson process.

The seller influences market demand by dynamically adjusting its prices, p_t, over the time interval $t \in [0, T]$. When a potential customer arrives, the customer decides whether to make a purchase based on the current price. Given a price of p, the potential customer decides to buy the product with probability $\lambda(p)$. Thus, the purchase rate of the product can be expressed as follows:

$$\Lambda_t(p) := \lambda(p)\left(\varphi + h\left(\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}\right)\right). \tag{1}$$

For each unit of sales, the seller incurs a marginal cost of c. Following the convention in RM (Gallego and Van Ryzin 1994), we use the purchase probability (also known as the demand rate) λ as the decision variable and make the following standard assumptions.

Assumption R (Regularity Assumption). *The purchase probability $\lambda(p) \geq 0$ is differentiable and invertible in p; The profit rate $r(\lambda) = \lambda(p(\lambda) - c)$ is differentiable, bounded, and strictly concave in λ.*

Denote $[\underline{\lambda}, \overline{\lambda}]$ as the range of feasible purchase probabilities. The seller selects a pricing policy $u = \{\lambda_t(\mathcal{H}_t) \in [\underline{\lambda}, \overline{\lambda}] : t \in [0, T]\}$, where $\mathcal{H}_t$ is the history prior to t, to maximize the expected profit from t onward

$$V_t(\{\mathcal{H}_t\}_{s\leq t}) := \max_u \mathbb{E}_u \int_t^T (p(\lambda_s) - c) dN_s, \tag{2}$$

where N_t is a general Hawkes process whose rate is given by Equation (1).

3.2. Structure of Optimal Policy

Before presenting the structure of the optimal policy, we define the excitement level

$$x_t := \sum_{t_i<t} \alpha e^{-\beta(t-t_i)}.$$

Here, x_t is the cumulative influence intensity of earlier buyers, which is stimulated by an immediate increase after a purchase and gradually decays over time between two consecutive purchases. The dynamic pricing system can be represented with the single state variable x_t. It is sufficient to show that $\lambda_t(\mathcal{H}_t)$'s dependence of $\mathcal{H}_t$ is only through x_t. Hence, we write $\lambda_t(x_t)$ onward.

In the following lemma, we characterize the dynamics of x_t with a stochastic differential equation.

Lemma 1. *The excitement level x_t is characterized by the following differential equation with stochastic jumps:*

$$dx_t = -\beta x_t dt + \alpha dN_t,$$

where N_t is a point process with intensity $\Lambda_t(p) = \lambda_t(\varphi + h(x_t))$, and $x_0 = 0$.

Lemma 1 characterizes the excitement level x_t, which increases by α after every purchase and decays over time at the rate of β between two consecutive purchases. Next, we characterize the optimal expected profit function, which depends on the intensity of potential customers. Different from the constant arrival intensity in the traditional RM problem, the potential customer intensity in our setting is nonstationary. This intensity is a function of arrival times of all previous purchases, $\{t_i : i < N_t\}$. Tracking the timing of each purchase would significantly increase the dimensionality. By defining the excitement level, we can simplify potential customers' intensity and present the Hamilton-Jacobi-Bellman (HJB) condition in the following proposition.

Proposition 1 (HJB Equation). *The optimal expected profit $V_t(x) = V_t(\{N_s\}_{s\leq t})$ is determined by the following differential equation:*

$$-\frac{\partial}{\partial t} V_t(x) + x\beta \frac{\partial}{\partial x} V_t(x) = (h(x) + \varphi) \max_\lambda \{r(\lambda) + \lambda(V_t(x+\alpha) - V_t(x))\}, \tag{3}$$

with boundary condition $V_T(x) = 0$ for any $x \geq 0$.

On the right-hand side of the above equation, $V_t(x+\alpha) - V_t(x)$ can be viewed as the future impact of one additional purchase. The optimal intensity is determined to balance the current period profit and the future profit lift. We denote $\lambda_t^*(x)$ as the unique solution to Equation (3) and examine the structural properties of the optimal policy under Assumption R assuming that the profit rate function is strictly concave. Although Equation (3) in general does not have closed-form solutions, we illustrate the optimal solution through the following special case.

Example 1. When $\beta = 0$, $c = 0$, $h(x) = x$, and the demand function $\lambda(p) = ae^{-p}$:

$$\lambda_t^*(x) = \frac{a}{\alpha a(t-T) + e}, \quad p_t^*(x) = \ln(\alpha a(t-T) + e).$$

To ensure that the expression is well defined, we let $e > a\alpha T$. The optimal expected profit is given by

$$V_t(x) = \frac{1}{\alpha}(1 - \ln(\alpha a(t-T) + e))x - \frac{\varphi}{\alpha}(\ln(\alpha a(t-T) + e) - 1).$$

In this special case, the optimal purchase probability and price are independent of the excitement level. This finding is intriguing and somewhat counterintuitive. It suggests that, regardless of the number of

earlier buyers, the optimal purchase probability and price depend only on the remaining time in the horizon. We further investigate the linear self-exciting function in the following lemma.

Lemma 2 (Linear Self-Exciting Function). *If $h(x) = h_1 x + h_0$ for $h_1 \geq 0$ and $h_0 \geq 0$, the value function $V_t(x) = \mu_t x + \omega_t$, where μ_t and ω_t satisfy the following equations:*

$$\frac{d\mu_t}{dt} = \beta\mu_t - h_1 \max_{\lambda}\{r(\lambda) + \lambda\alpha\mu_t\},$$
$$\frac{d\mu_t}{dt} = \beta\mu_t + \frac{h_1}{h_0+\varphi}\frac{d\omega_t}{dt}, \tag{4}$$

with $\mu_T = 0$ and $\omega_T = 0$. The optimal purchase probability $\lambda_t^* = \arg\max_{\lambda\in[\underline{\lambda},\overline{\lambda}]}\{r(\lambda) + \lambda\alpha\mu_t\}$, which is independent of x.

With linear self-excitement, a higher level of excitement proportionally increases the influenced customer's arrival rate, and hence the change in the optimal purchase rate is independent of the current level of excitement. Consequently, the seller's optimal pricing decision, which balances immediate profit with future benefits, also remains independent of the current excitement level. This implies that pricing managers can implement a predictable open-loop pricing trajectory rather than contingent adaptations based on the level of excitement. Such a pricing policy not only reduces the complexity of pricing decisions but also allows managers to focus on consistent marketing activities over the sales horizon rather than on the number of users making purchases.

In the linear self-excitement case, the monotonicity of the optimal pricing decisions can be established quite straightforwardly. However, when self-excitement is nonlinear, the value function is no longer linear. Instead, it is shaped by the properties of the self-exciting function itself. As a result, existing analytical methods used to establish monotonicity under linear self-excitement cannot be directly extended to the nonlinear setting. The shape of the self-exciting function, whether convex or concave, also influences the pricing decision and warrants a more in-depth analysis. Next, we consider the simplest benchmark case of the pricing policy without self-excitement. When $\alpha = 0$ or $\beta \to \infty$, the customer arrival process is reduced to a time-homogeneous Poisson process. We have the following observation.

Observation 1 (No Self-Excitement Benchmark). If $\alpha = 0$ or $\beta \to \infty$, customers arrive following a Poisson process, and the optimal intensity is constant, that is, $\lambda_t^* = \lambda^0 = \arg\max_{\lambda\in[\underline{\lambda},\overline{\lambda}]}\{r(\lambda)\}$.

Without self-excitement, the optimal intensity (price) remains constant and is set to maximize the profit rate $r(\lambda)$, which is the one for Poisson arrivals. However, as mentioned, with self-excitement, the term $V_t(x+\alpha) - V_t(x)$ on the right-hand side of Equation (3) represents the future impact of an additional purchase. With this future effect taken into account, the optimal policy generally does not myopically maximize the immediate profit rate $r(\lambda)$. Instead, it balances immediate profit with the future benefit of increased excitement. The following proposition shows how the optimal expected profit function varies with the time, the excitement level, and the parameters of self-exciting process.

Proposition 2 (Monotonicity Properties of Optimal Expected Profit Function).

(i) *The optimal expected profit function $V_t(x)$ is decreasing in t and strictly increasing in x;*

(ii) *The optimal expected profit function $V_t(x)$ is strictly increasing in α and φ, and strictly decreasing in β.*

With everything else equal, from Proposition 2(i), the optimal expected profit-to-go decreases as time elapses and gets closer to the end of the sales horizon. Moreover, the optimal expected profit-to-go increases with the current excitement level. Proposition 2(ii) demonstrates that an increase in the excitation intensity α or the organic arrival rate φ attracts additional customers (in a stochastic sense), thereby increasing the optimal expected profit. Similarly, a decrease in the decay rate β leads to more customer arrivals (in a stochastic sense) and, consequently, increases the optimal expected profit. A direct implication of these monotonicity properties is that the optimal expected profit under the self-exciting process is larger than that under the Poisson arrivals (without self-excitement). Additionally, the intensity of self-exciting arrivals exceeds that of the Poisson process. We now proceed to examine the properties of the optimal purchase rate.

Proposition 3. *The optimal purchase rate $\Lambda_t^*(x) := \lambda_t^*(x)(\varphi + h(x))$ increases in x.*

Proposition 3 states that, under the seller's optimal pricing policy, the purchase rate increases with the excitement level. This optimal policy consistently encourages additional purchases as the influence of earlier buyers grows. Notably, this proposition holds without any specific assumptions about the self-exciting function $h(x)$. Later, in Theorem 2, we find that the optimal purchase probability $\lambda_t^*(x)$ and the optimal price can either increase or decrease in the excitement level depending on the shape of $h(\cdot)$. However, we find that the optimal purchase rate always increases with the excitement level, even if the seller increases prices for a higher excitement level. The following theorem demonstrates the monotonicity of the optimal solutions with respect to the elapsed time t.

Theorem 1 (Static Monotonicity in t). *For any $x \geq 0$, $\lambda_t^*(x)$ and $\Lambda_t^*(x)$ decrease in t, and $p_t^*(x)$ increases in t.*

Theorem 1 indicates that both the optimal purchase probability and the optimal purchase rate decline with

the elapsed time, which implies that the optimal price correspondingly increases with the elapsed time. The latter means that for the same level of excitement, at the beginning of the sales horizon, the seller sets a lower price to attract more buyers, thereby generating additional potential arrivals in the future, whereas as the end of the sales horizon approaches, the seller raises the price to gain immediate profits. The driving force behind this is that the same excitement level has a higher value in driving future sales at the earlier horizon than at the later.

In Proposition 3, we demonstrate that the optimal purchase rate increases with the excitement level x. The purchase rate is determined by the purchase probability $\lambda_t^*(x)$, which is set by the seller, and the customer arrival intensity $\varphi + h(x)$. Because the latter increases with the excitement level x, the monotonicity of the optimal purchase probability in the excitement level is not immediately evident from Proposition 3. To address this, we establish the monotonicity of the optimal purchase probability $\lambda_t^*(x)$ in the following theorem and explore the convexity and concavity of the optimal expected profit function $V_t(x)$ with respect to the excitement level x.

Theorem 2 (Static Monotonicity in x). *For any $t \in [0, T)$,*

(i) *If $h(x)$ is strictly convex, $V_t(x)$ is strictly convex in x, and $\lambda_t^*(x)$ (respectively, $p_t^*(x)$) strictly increases (respectively, decreases) in x.*

(ii) *If $h(x)$ is strictly concave, $V_t(x)$ is strictly concave in x, and $\lambda_t^*(x)$ (respectively, $p_t^*(x)$) strictly decreases (respectively, increases) in x.*

(iii) *If $h(x)$ is linear, $V_t(x)$ is linear in x, and $\lambda_t^*(x)$ (hence, $p_t^*(x)$) is independent of x.*[3]

The optimal policy balances maximizing immediate profit and the pool of potential customers for future benefits. On the one hand, when the seller lowers the purchase probability (equivalently, increases the price), the total immediate purchase rate decreases. This results in a gradual reduction in the excitement level x and, consequently, a decrease in the number of influenced customers (in a stochastic sense), which lowers expected future profits. On the other hand, increasing the price also immediately increases the profit rate. Theorem 2 demonstrates that the optimal purchase probability increases (respectively, decreases) with respect to the excitement level x if and only if the self-exciting function is convex (respectively, concave). As the level of excitement increases, one might expect the seller to raise the price to maximize profit. However, we find that the optimal price may decrease as excitement levels increase.

A convex self-exciting function $h(x)$ can correspond to a stage of rapid market growth for a product. In this phase, a small increase in the excitement level x results in a significant expansion of influenced customers (in a stochastic sense), which yields substantial expected future profits. Therefore, when the excitement level x is higher, the seller further reduces the price to attract a larger pool of buyers to leverage their influence.

Conversely, a concave self-exciting function $h(x)$ can reflect a phase of market saturation for a product. In this phase, the pool of potential customers is limited, and the benefit of increasing the excitement level x diminishes. In this case, the optimal purchase probability (respectively, price) decreases (respectively, increases) with the excitement level, leading to an immediate increase in the profit rate while yielding limited expected future profits. Thus, when the excitement level x is higher, the seller increases the price to generate a higher immediate profit and exploit the higher excitement level.

Pricing managers should regularly analyze market conditions to determine whether the market is in a growth or saturation phase. In a growth phase, engaging early adopters who are likely to spread positive word-of-mouth is crucial; using incentives, discounts, or loyalty programs can encourage sharing experiences and attract more customers. Conversely, in a saturation phase, when the existing customer base is larger, the seller can exploit it by charging higher prices. By following these strategies, pricing managers can effectively leverage the dynamic interplay between pricing and customer engagement to optimally balance immediate and future profits.

3.3. Optimal Price Trend

From Theorems 1 and 2, we characterize the monotonicity properties of the optimal price with respect to time t and the excitement level x. In the following theorem, we investigate the (stochastic) price trend over time by studying the average of the sample paths of the optimal price. Denote by $g(p) = -\lambda'(p)/\lambda(p)$ the hazard rate of the customer valuation distribution and by $1/g(p)$ the Mills ratio.

Theorem 3 (Optimal Price Trend). *Assume that $1/g(p)$ is convex in p and $\beta = 0$. $\mathbb{E}[p_s^*(x_s) \mid \mathcal{F}_t] \geq p_t^*(x_t)$ for any $0 \leq t < s \leq T$.*

The condition that $1/g(p)$ is convex in p is satisfied by many common distributions such as normal, uniform, logistic, Weibull, and gamma with shape parameters no less than one (Xu and Hopp 2009, Pang et al. 2015). Theorem 3 shows that under the special case where the excitement level x_t does not decay over time (i.e., $\beta = 0$), the optimal price process has an upward trend.

The pattern of increasing prices over time aligns with the common practice of managers offering early-bird discounts at the beginning of the sales horizon and using phrases like "limited time offer" or "today only" to attract a larger customer base rapidly. As the sales period progresses and early sales stimulate later arrivals through social effects, gradually increasing prices

can later capitalize on the large customer base established in the early stage. Furthermore, this upward price trend provides managers with a guideline for evaluating their pricing policy. If prices gradually decrease throughout the day, it suggests that the implemented pricing policy may not be profitable.

4. Asymptotic Analysis

We have shown the structural properties of the optimal policy in the previous section. Because there is no general closed-form solution for the optimal intensity as a function of the state variable, a common approach to numerically calculate the optimal solution is to discretize t with a small step size $\delta > 0$ and compute $V_t(x)$ and $\lambda_t(x)$ backward from the end of the horizon T. However, such a calculation would require the value of the state, $x+\alpha$, at time $t+\delta$. Thus, if we would like to find the optimal solution $\lambda_t(x)$ for any $t \geq 0$ and $x \leq \bar{x}$, we must iteratively calculate all of the value functions $\{V_t(x): \ x \leq \bar{x} + \lceil \alpha t/\delta \rceil\}$. Significant computational effort may be required to compute the optimal intensity, especially for large-scale problems.

We generalize the formulation of the original problem to study the heuristics and their performance for large-scale problems. Motivated by prior works in RM such as Gallego and Van Ryzin (1997), Cooper (2002), and Maglaras and Meissner (2006), we consider a series of problems: in the nth problem, the Hawkes process N_t admits the intensity $\Lambda_t^{(n)} = n\lambda_t[\varphi + h(x_{t-}/n)]$. The system state x_t is given by $dx_t = -\beta x_t dt + \alpha dN_t$ and $x_0 = 0$. The terminal payoff at $t = T$ is $V_T^{(n)}(x) = 0$ for any x. The problem we studied in the previous sections corresponds to the problem where $n = 1$. We first provide asymptotic bounds of the expectation and variance of x_t with respect to n.

4.1. Asymptotic Bounds of $\mathbb{E}[x_t]$ and $\mathrm{Var}[x_t]$

As we study the problems where n is large, we would like to ensure that the problems are "properly" scaled: One would expect $\mathbb{E}[x_t] = O(n)$ as we scale Λ_t by n. However, this seemingly obvious property may not hold for general $h(\cdot)$ because of the self-exciting property. In particular, because x_t is unbounded, it is possible to have a small subset of sample paths where x_t/n is very large. For certain convex $h(x)$, it is possible that such a small number of cases drive $\mathbb{E}[x_t]/n \to \infty$. In the following proposition, we provide a sufficient condition that ensures $\mathbb{E}[x_t]$ is bounded by n.

Proposition 4. *If $h(x)$ is Lipschitz continuous, $\mathbb{E}[x_t^u] = O(n)$ for any Markov policy u.*

The Lipschitz condition in Proposition 4 ensures that the arrival rate $\Lambda_t^{(n)}$ does not increase "too fast." It is worth noting that the Lipschitz condition is also shown as a sufficient condition for the stability of the nonlinear Hawkes process in Brémaud and Massoulié (1996). It guarantees that the average intensity is finite over any time interval. Therefore, in the asymptotic analysis, we make the following additional assumption about $h(x)$.

Assumption LD. *The self-exciting function $h(\cdot)$ is Lipschitz continuous and infinitely differentiable.*

From Theorem 2, the optimal intensity λ_t^* is dependent on the excitement level x_t, which follows a stochastic process. Consequently, λ_t^* also varies at time t depending on the prior demand realizations. As a result, the variance of N_t^* and x_t^* under the optimal purchase probability may be even higher than that of a Hawkes process with a deterministic intensity. Nonetheless, in the following proposition, we establish the asymptotic bounds of the variance of λ_t^*, N_t^*, and x_t^*.

Proposition 5. *Suppose Assumption LD holds. Under the optimal policy,*

(i) $\mathrm{Var}[\lambda_t^*] = O(1/n)$, $\mathrm{Var}[N_t^*] = \Theta(n)$, *and* $\mathrm{Var}[x_t^*] = \Theta(n)$.

(ii) *If $|h''(x)| > 0$ for any $x \geq 0$,* $\mathrm{Var}[\lambda_t^*] = \Theta(1/n)$.

As established in Daw and Pender (2018), the variance of N_t, the number of arrivals at time t in the Hawkes process, is larger than that of the standard Poisson process due to the self-exciting property. Here we show in Proposition 5 that the variance of N_t^* is of the same magnitude as n as long as the variance of the optimal purchase intensity is not too large. In addition, the magnitude of $\mathrm{Var}[\lambda_t^*]$ is upper bounded by $1/n$, which can be achieved if the curvature of $h(x)$ is nonzero for any x.

Note that the results of Proposition 5 are more general than those in the existing literature. Gao and Zhu (2018) derive a central limit theorem for the linear Hawkes process using the immigration-birth representation. In the linear setting, Theorem 2 implies that the optimal intensity $\lambda_t^*(x)$ is independent of x, yielding zero variance and making the result in Proposition 5 immediate. In contrast, we consider a general nonlinear Hawkes process with endogenously controlled intensity via a dynamic purchase probability. These features introduce greater analytical challenges.

Proposition 5 may not hold for a self-exciting function $h(x)$ that is not Lipschitz continuous. For a general function that is infinitely differentiable, we can construct a corresponding Lipschitz continuous function $\hat{h}(x)$ that approximates $h(x)$ by a linear extrapolation, that is, with the tangent line when x is sufficiently large:

$$\hat{h}(x) = \begin{cases} h(x) & \text{if } x \leq \bar{x}, \\ h(\bar{x}) + h'(\bar{x})(x - \bar{x}) & \text{if } x > \bar{x}. \end{cases}$$

The constructed function $\hat{h}(x)$ has a bounded first derivative, so it is Lipschitz continuous. By Proposition 5, if we choose $\bar{x} \geq 1 + \mathbb{E}[x_t^*]/n$, $\mathbb{P}(x_t^*/n \geq \bar{x}) \leq \mathbb{P}(x_t^* \geq n + \mathbb{E}[x_t^*]) \leq \mathrm{Var}[x_t^*]/n^2 = O(1/n)$ where the second

inequality is due to Chebyshev's inequality. Such an approximation of $h(\cdot)$ only affects a very small set of cases. Thus, our Lipschitz continuity assumption is not overly restrictive.

4.2. Deterministic Heuristic and Profit Bounds

In a traditional stochastic RM problem, a popular heuristic is adopted from the deterministic version, in which customers arrive in a fluid fashion. Gallego and Van Ryzin (1994) has shown that the deterministic heuristic is asymptotically optimal when the scale of the problem $n \to \infty$. The authors show that the revenue loss of adopting the deterministic heuristic is at most on the scale of $\sqrt{n}$ relative to the revenue of the deterministic problem as a benchmark. To improve the revenue gap, reoptimization heuristics are proposed to resolve the deterministic problem and update the sales rate at distinct times during the sales horizon. Jasin (2014) shows that the heuristic that resolves the deterministic problem at discrete time points $t = 1, 2, \ldots, n$ achieves a $\log(n)$ performance gap when compared with the revenue of the deterministic problem. Moreover, the author derives an alternative heuristic that resolves more frequently when it gets closer to the end and shows that such a heuristic achieves the same $\log(n)$ performance bound.

Unlike traditional RM problems, the state variable x_t is not capped in our problem. It can increase rapidly due to the self-exciting property. Nonetheless, in search for a well-performing heuristic, we also take a similar approach and investigate the deterministic problem where customers arrive in a fluid fashion with rate $\Lambda_t = n\lambda_t(\varphi + h(x_t/n))$. Here, x_t, unlike in the stochastic problem, is calculated deterministically for a given λ_t and x_t:

$$\begin{aligned} x_{t+\delta} &= (1 - \Lambda_t\delta)(x_t - \beta x_t\delta) + \Lambda_t\delta(x_t + \alpha - \beta x_t\delta) \\ &= x_t + (\alpha\Lambda_t - \beta x_t)\delta = x_t + [\alpha n\lambda_t(\varphi + h(x_t/n)) - \beta x_t]\delta. \end{aligned}$$

Rearranging the terms and taking $\delta \to 0$, we have $dx_t/dt = \alpha n\lambda_t(\varphi + h(x_t/n)) - \beta x_t$. Therefore, the deterministic version of our problem can be formulated as the following optimal control problem:

$$\begin{aligned} V^d(n) = \max_{\lambda_t} \quad & \Pi^d(\lambda_t) = \int_0^T n\lambda_t\left[\varphi + h\left(\frac{x_t}{n}\right)\right] \cdot (p(\lambda_t) - c)dt \\ \text{s.t.} \quad & \frac{dx_t}{dt} = \alpha n\lambda_t\left[\varphi + h\left(\frac{x_t}{n}\right)\right] - \beta x_t, \quad x_0 = 0. \end{aligned}$$

In a traditional RM problem, the optimal deterministic arrival rate is fixed throughout the sales horizon. However, in our problem, the optimal rate varies in time t. Nevertheless, there exist numerical techniques (e.g., forward-backward sweep method) that allow us to calculate numerical solutions efficiently. We establish the connection between the stochastic problem and the corresponding deterministic problem. First, we consider the special case where the self-exciting function $h(\cdot)$ is linear.

Proposition 6. *If the self-exciting function $h(\cdot)$ is linear, $V^*(n) = V^d(n)$ and $\lambda_t^* = \lambda_t^d$.*

Proposition 6 follows from two results: When $h(\cdot)$ is linear, λ_t^* is independent of x_t according to Theorem 2(iii), and $\{\mathbb{E}[x_t^*]\}$ has the same dynamics as in the deterministic problem. This means that using λ_t^* in the deterministic problem will give the same profit as the expected profit in the stochastic problem. Using λ_t^d in the stochastic problem also yields the same expected profit as the optimal profit in the deterministic problem. Thus, the stochastic and deterministic problems are one and the same when $h(\cdot)$ is linear.

In general, when $h(\cdot)$ is not linear, the above arguments do not hold exactly. Nonetheless, we demonstrate in Proposition 5 that the variance of the optimal policy λ_t^*, and the corresponding x_t^* are "small." Thus, using similar logic as discussed above, we apply the expected intensity $\{\mathbb{E}[\lambda_t^u]\}$ of a policy λ_t^u to the deterministic problem, and bound the difference in profits of the two corresponding problems.

Proposition 7. *Suppose Assumption LD holds. For any Markov policy u such that* $\mathrm{Var}[\lambda_t^u] = O(1/n)$,

$$V^u(n) = \Pi^d(\{\mathbb{E}[\lambda_t^u]\}) + O(1).$$

Proposition 7 states that for a subclass of the policies where the variance of the intensity λ_t^u is small, the difference between the expected profit $V^u(n)$ in the stochastic problem and the profit of using $\{\mathbb{E}[\lambda_t^u]\}$ in the deterministic problem is bounded by a constant.

We now study the performance of the deterministic heuristic policy DH, where we adopt the optimal solution in the deterministic problem λ_t^d as the deterministic sales rate in the stochastic problem. In the following theorem, we bound the optimal expected profit $V^*(n)$ and the expected profit of the deterministic heuristic $V^{DH}(n)$ in relation to the optimal profit $V^d(n)$ in the deterministic problem.

Theorem 4 (Performance Gap). *Suppose Assumption LD holds. There exist two constants K_1 and K_2 independent of n, such that $|V^{DH}(n) - V^d(n)| \le K_1$ and $|V^*(n) - V^d(n)| \le K_2$.*

Theorem 4 is established by applying Proposition 7 to both the deterministic heuristic and the optimal policy. The critical assumption required by Proposition 7, that $\mathrm{Var}[\lambda_t^u] = O(1/n)$ for all t, is satisfied by these two policies. However, we note that the static heuristic in the traditional RM problem with an inventory constraint may not satisfy such a condition: Even though the sales rate is fixed under the static heuristic, the probability of sellout increases as it gets closer to the end of the sales horizon, which means that $\mathrm{Var}[\lambda_t^{DH}] = \Theta(1)$ as $t \to T$.

When the inventory constraint under the optimal solution is tight in the deterministic problem, Wang and Wang (2022) show that $V^d(n) - V^*(n) = \Theta(\ln(n))$. Gallego and Van Ryzin (1994) derive the exact bound of $V^d(n) - V^{DH}(n) = \Theta(\sqrt{n})$. When the inventory constraint is not tight, Wang and Wang (2022) show that $V^d(n) - V^*(n) = O(1)$. Although not directly shown in Gallego and Van Ryzin (1994), it can be easily derived that $V^d(n) - V^{DH}(n) = O(1)$. In our problem, we demonstrate in Theorem 4 that both $|V^*(n) - V^d(n)|$ and $|V^{DH}(n) - V^d(n)|$ are bounded by a constant. As a direct implication, the performance loss of the deterministic heuristic is also bounded by a constant, that is, $V^{DH}(n) - V^*(n) = O(1)$. For large-scale problems, the deterministic heuristic is not only asymptotically optimal but also incurs negligible performance loss. Because the performance loss of deterministic heuristics is bounded by a constant and becomes negligible for large-scale problems, pricing managers can confidently use these heuristics for large-scale decision making without concern for efficiency loss. The superior performance of deterministic heuristics will be numerically tested and confirmed in Section 6.

5. Model Extensions

In this section, we extend our base model to account for discrete pricing, immigration-birth process, and multiple products. We explain how the earlier results can be applied to these new contexts.

5.1. Discrete Price Case

In this section, we assume that the seller offers a finite set of exogenously determined discrete prices denoted by $\{p_1, p_2, \ldots, p_K\}$ with $p_1 < p_2 < \cdots < p_K$ and corresponding purchase probabilities $\lambda_1 > \lambda_2 > \cdots > \lambda_K$. Given that the optimal purchase probability is greater than the Poisson arrival intensity λ^0, we assume that $\lambda_1 > \lambda_2 > \cdots > \lambda_K \geq \lambda^0$. The objective function is updated as follows:

$$-\frac{\partial}{\partial t} V_t(x) + x\beta \frac{\partial}{\partial x} V_t(x) = (h(x) + \varphi) \max_{\lambda \in \{\lambda_1, \ldots, \lambda_K\}} \{r(\lambda) + \lambda(V_t(x+\alpha) - V_t(x))\}.$$

It can be easily verified that the value function $V_t(x)$ satisfies parts (i) and (ii) of Proposition 2 and Theorem 2(i). Furthermore, we can derive the optimal intensity in the following theorem.

Theorem 5. *For $t \in [0, T]$,*

(i) *If $h(x)$ is convex, there exists $0 \leq R_{t,1} \leq R_{t,2} \leq \cdots \leq R_{t,K-1}$, such that $\lambda_t^*(x) = \lambda_1$ if $x < R_{t,1}$; $\lambda_t^*(x) = \lambda_i$ if $R_{t,i-1} \leq x < R_{t,i}$, for $i = 2, \ldots, K-1$; $\lambda_t^*(x) = \lambda_K$, otherwise.*

(ii) *If $h(x)$ is linear, there exists $0 \leq \hat{T}_1 \leq \hat{T}_2 \leq \cdots \leq \hat{T}_{K-1} \leq T$, such that $\lambda_t^*(x) = \lambda_1$ if $t < \hat{T}_1$; $\lambda_t^*(x) = \lambda_i$ if $\hat{T}_{i-1} \leq t < \hat{T}_i$, for $i = 2, \ldots, K-1$; $\lambda_t^*(x) = \lambda_K$, otherwise.*

Theorem 5 demonstrates that, on the one hand, with a linear self-exciting function, the optimal intensity is constant within certain period regardless of the excitement level and increases across periods. On the other hand, with a convex self-exciting function, there are $K-1$ thresholds on the excitement level at each time point, resulting in the optimal intensity being the same when the excitement level lies between two consecutive thresholds. This raises a pertinent question: can the discrete price heuristic be asymptotically optimal? Somewhat surprisingly, we will show that the finitely discrete price heuristic is not asymptotically optimal in the following theorem.

Theorem 6. *Suppose Assumption LD holds. Consider any heuristic u where the available price set is finite. The heuristic is not asymptotically optimal, that is, $V^*(n) - V^u(n) = \Theta(n)$.*

Theorem 6 demonstrates that the discrete price heuristic is not asymptotically optimal. This finding is both intriguing and distinct from the RM literature. From Theorem 4, for any heuristic to be asymptotically optimal, the expected purchase probability and excitement level must be "close" to the optimal solution in the deterministic problem. Unlike traditional RM problems, in which the solution is time invariant, the optimal deterministic solution in our problem is time varying. Even though it may be possible to match the expected excitement level with discrete prices, doing so would lead to suboptimal profit due to the concavity of the profit rate function $r(\cdot)$. Because the discrete price heuristic is not asymptotically optimal, pricing managers had better avoid relying solely on a fixed set of discrete prices. Instead, they are better off constantly monitoring market trends and customer behavior and updating prices accordingly without confining themselves to the given discrete price set. This responsiveness to market changes will help set prices that are more closely aligned with the optimal pricing policy, without the constraint of selecting from a predetermined set of discrete prices.

Next, we consider the case in which the seller first selects a set of K discrete price sets and is then restricted to choosing prices from this set. Let $V^*(n, K)$ denote the optimal expected profit under system scale n when at most K discrete price levels are available. We characterize the optimality gap as follows.

Proposition 8. *If $h(x)$ is linear, the optimality gap satisfies $V^*(n) - V^*(n, K) = O(n/K)$.*

Proposition 8 demonstrates that the loss in the optimal expected profit due to price discretization is of order $O(n/K)$ and decreases as the number of available price levels K increases. This result arises because the optimal purchasing probability varies continuously and can be better approximated with more

discretizations. As the discretization becomes more granular, the seller gains greater flexibility in synthesizing the optimal policy using discrete prices, thereby reducing the gap in the expected profit.

5.2. Immigration-Birth Process

In this section, we establish the connection between the self-exciting process with $h(x) = x$ and the immigration-birth process. We begin by defining the general immigration-birth process, where immigrants arrive according to a Poisson process with rate φ. Each immigrant produces descendants at time $t \geq t_i$ with a rate $\alpha e^{-\beta(t-t_i)}$. Consequently, at any epoch t, the rate of generation 1 arrivals, accounting for both internal births and external immigration, is given by $\varphi + \sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$.

We further assume that individuals, whether immigrants or their descendants, who are present at epoch t decide whether to remain in the system with a probability of $\lambda_t^*(x)$, which corresponds to the optimal arrival rate in our baseline model. From Theorem 2, if $h(x)$ is linear, $\lambda_t^*(x)$ is independent of the excitement level x. Combining this with Theorem 1, the probability $\lambda_t^*(x)$ of remaining in the system is independent of x and decreases over time. Therefore, $\lambda_t^*(x)$ can be simplified to λ_t, a function that solely depends on t and decreases as t increases. Thus, the overall population generation rate in this system, which includes contributions from both internal births and external immigration, can be expressed as follows:

$$\Lambda_t := \lambda_t \left(\varphi + \sum_{t_i<t} \alpha e^{-\beta(t-t_i)} \right). \tag{5}$$

We now demonstrate that the general self-exciting process with linear excitement is equivalent to the immigration-birth process defined above. Comparing the generation rate with the arrival rate in (1) for $h(x) = x$, it becomes evident that the arrival rates in both processes are identical. Therefore, by the only lemma in Hawkes and Oakes (1974), the general self-exciting process in (1) and the immigration-birth process in (5) represent the same point process.

However, our model differs from the classic immigration-birth process due to the introduction of the dynamic purchase probability. The classic process without control, as studied in Oakes (1975), assumes that when a potential customer arrives, the customer produces offspring with probability $\varphi + \sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$, where the descendants of any arrival follow the same distribution, which determines the total number of descendants and the mean of the population. In contrast, in our model, the number of descendants up to time t by an immigrant follows a nonstationary Poisson distribution with mean given by

$$m_t = \int_0^\infty \lambda_{u+t} \alpha e^{-\beta u} du. \tag{6}$$

As λ_{u+t} decreases over time, the number of descendants is no longer constant. To excite more customers in the future, λ_t is lower bounded by the Poisson arrival rate λ^0. Additionally, because λ_t represents a purchasing probability, it is upper-bounded by one. We thus have that as time progresses, the expected number of descendants converges to a value in the range $[\lambda^0\alpha/\beta, \alpha/\beta]$. Returning to the general Hawkes process, this implies that the future profits from a single purchasing organic customer are bounded and converge over time. Similarly, because λ_t is a decreasing function that remains greater than zero, the purchasing probability λ_t converges. We now derive the total number of descendants using the above arguments.

Proposition 9 (Convergence of Overall Population Generation Rate). *The number of descendants for each immigrant converges to* $m = \frac{\alpha/\beta}{\varphi/\lambda^0+\alpha/\beta}$. *The stationary purchasing probability is given by* $\lambda = \frac{1}{\varphi/\lambda^0+\alpha/\beta}$. *The total number of descendants by any immigrant is finite with probability* 1. *The overall population generation rate* $\Lambda_t \to \lambda\varphi/(1-m)$, as $t \to \infty$.

Proposition 9 ensures that the number of descendants for any immigrant remains finite with probability 1 under the optimal policy, which prevents low-margin runaway growth. The key insight is that pricing decisions can be profitably leveraged to control population expansion. By optimizing pricing strategies, firms can ensure that each immigrant produces a manageable number of descendants, thereby avoiding a population that grows uncontrollably, as seen in classic immigration and birth systems without pricing. This allows managers to effectively control the self-exciting process, preventing a situation in which population growth becomes explosive on a thin margin, which could lead to profit loss and operational challenges.

5.3. Multiple Products

In practice, a seller may offer a number of products in the same category that only differ in size and style. One purchase of any product can excite future customers who may go on to purchase other products in the same category. In this section, we assume that there are M differentiated products. A common customer stream is shared by all products, with the customer arrival rate at time t modeled as a multivariate stochastic point process. Specifically, we assume that it follows a multivariate self-exciting process, defined as follows: $\Phi_t(\mathcal{H}_t) = \varphi + h\left(\sum_{n=1}^M \sum_{t_i^n<t} \alpha e^{-\beta(t-t_i^n)}\right)$, where $\alpha \geq 0$, $\beta \geq 0$, $\varphi > 0$, and $h(\cdot)$ is a positive and increasing function. The price vector of those products is denoted by $\mathbf{p} = (p^1, p^2, \ldots, p^M)$. Upon a customer arrival, the customer's purchasing probability of product $m = 1, 2, \ldots, M$ is denoted by $\lambda_t^m(\mathbf{p})$, determined by the

current price vector $\mathbf{p}$. For each unit of sales of product $m = 1, 2, \ldots, M$, the seller incurs a marginal cost of c^m. We thereafter write down the purchase rate as follows: $\Lambda_t^m(\mathbf{p}) = \lambda_t^m(\mathbf{p})\big(\varphi + h\big(\sum_{n=1}^M \sum_{t_i^n < t} \alpha e^{-\beta(t-t_i^n)}\big)\big)$. Following the conventions in RM (Gallego and Van Ryzin 1994), we use the purchase probabilities $\boldsymbol{\lambda} = (\lambda^1, \lambda^2, \ldots, \lambda^M)$ as the decision variables and make similar standard assumptions on the profit rate functions $r^m(\boldsymbol{\lambda}) = (p^m(\boldsymbol{\lambda}) - c^m)\lambda^m$ as in our base model.

By defining $x := \sum_{n=1}^M \sum_{t_i^n < t} \alpha e^{-\beta(t-t_i^n)}$ and $r(\boldsymbol{\lambda}) = \sum_{m=1}^M r^m(\boldsymbol{\lambda})$, following the same steps as in Proposition 1, we can express the seller's optimization problem as follows:

$$-\frac{\partial}{\partial t} V_t(x) + x\beta \frac{\partial}{\partial x} V_t(x) = (h(x) + \varphi) \max_{\boldsymbol{\lambda}_t = (\lambda_t^1, \lambda_t^2, \ldots, \lambda_t^M)} \left\{ r(\boldsymbol{\lambda}_t) + \sum_{m=1}^M \lambda_t^m (V_t(x+\alpha) - V_t(x)) \right\}. \tag{7}$$

Even with multiple products that contribute to exciting customers, the system dynamics are similar to those of a single product, as we can capture the system state by a single-dimensional exciting level x. Consequently, we expect that the seller's value function $V_t(x)$ has properties that align with Proposition 2. Moreover, by applying the first-order condition to the maximization problem on the right-hand side of (7), we obtain the following proposition.

Proposition 10 (Marginal Profits of Optimal Demand Rates). *At the optimal demand rates $\boldsymbol{\lambda}_t^*$, the marginal profit rates across products are equal, i.e., $\partial r(\boldsymbol{\lambda})/\partial \lambda^1|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}_t^*} = \partial r(\boldsymbol{\lambda})/\partial \lambda^2|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}_t^*} = \cdots = \partial r(\boldsymbol{\lambda})/\partial \lambda^M|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}_t^*}$.*

Proposition 10 demonstrates that at optimal demand rates, the marginal profit rates across products are equal. This result enables sellers to strategically adjust product prices so that the marginal profits for each product are equal. Additionally, this finding helps connect individual demand rates to the category's total demand rate, denoted as $\tilde{\lambda}_t = \sum_{m=1}^M \lambda_t^m$. By combining Proposition 10 with the definition of $\tilde{\lambda}_t$, we can express individual demand rates as functions of the total demand rate of the category, $\lambda_t^m(\tilde{\lambda}_t)$. Furthermore, by defining $r(\tilde{\lambda}) = r(\boldsymbol{\lambda}(\tilde{\lambda}))$, we can reformulate the optimization problem in Equation (7) as

$$-\frac{\partial}{\partial t} V_t(x) + x\beta \frac{\partial}{\partial x} V_t(x) = (h(x) + \varphi) \max_{\tilde{\lambda}_t} \{ r(\tilde{\lambda}_t) + \tilde{\lambda}_t (V_t(x+\alpha) - V_t(x)) \}, \tag{8}$$

which shows that, at the optimal demand rates, the problem simplifies to the single-product case as in our base model. Consequently, the results established in Theorems 1 and 2 remain valid. By solving this optimization problem, we can determine the optimal total demand rate for the category.

The individual demand rates can then be derived using $\lambda_t^m(\tilde{\lambda}_t)$. The relationship between a particular product's demand rate and the total category demand rate is captured by $\partial \lambda_t^m(\tilde{\lambda}_t)/\partial \tilde{\lambda}_t$. If $\partial \lambda_t^m(\tilde{\lambda}_t)/\partial \tilde{\lambda}_t > 0$, an increase in demand for an individual product m leads to an increase in the total demand for the category, thereby exciting the overall category's demand by attracting new customers. This case can be viewed as products being complements within the category. In this scenario, the seller may use such a product as a "loss leader," pricing it lower to boost demand for those with high margins within the category. Conversely, if $\partial \lambda_t^m(\tilde{\lambda}_t)/\partial \tilde{\lambda}_t < 0$, an increase in demand for the individual product m reduces the total demand for the category. This case can be viewed as products being substitutes within the category.

Although we omit the proofs due to repetition, all the asymptotic properties in Section 4 hold under the multiple products setup when we scale up the intensity by n. As the dynamics share a similar form with that of a single product in Equation (3), it can be shown that the optimal purchase probability for each product λ_t^m changes slowly at the rate of $1/n$ with respect to the excitement level x, the same rate as in our single-product base model. This indicates that the variances of λ_t^m are small, which serves as the basis of all asymptotic results.

6. Numerical Study

In this section, we propose various easy-to-implement heuristics for the base model to derive further insights into the seller's heuristic decision-making processes. Sections 6.1 and 6.2 evaluate the impact on profit margins of neglecting customers' self-exciting nature and implementing static pricing, respectively. Section 6.3 investigates the performance of the deterministic heuristics.

Our base numerical setting is essentially the same as that in our motivating example. We analyze sunscreen products on March 8 from JD.com. Using the maximum likelihood method, we identify a self-exciting process with parameters $\varphi = 0.392$, $\tilde{\alpha} = 0.09$, and $\beta = 0.11$. We assume that the customer valuations are scaled and uniformly distributed in [0, 1]. The organic arrival intensity of customers with independent needs is normalized to one, and the excitation intensity is calculated as $\alpha = \tilde{\alpha}/\varphi = 0.091/0.392 \approx 0.23$. Given that the item was sold over approximately 10 hours, we set the time horizon to 10 hours, that is, $T = 600$ minutes. We normalize the marginal cost to zero and adopt the self-exciting function as $h(x) = x + 0.001x^2$. To achieve convergence of the optimal solution in our computations, we use the semi-Lagrangian method to solve the HJB

equation. We discretize x with a step size of 0.01 and time t with a step size of 0.1. Linear interpolation is applied when the calculated values do not align with the lattice points. Unless otherwise specified, the parameters utilized in this section adhere to the aforementioned base numerical settings.

6.1. Profit Loss of Myopic Pricing

In this section, we examine the profit loss of myopically maximizing the profit rate for a given customer arrival and neglecting a customer's future impact on exciting other customers. When the seller ignores the self-excitement of the arrival process and considers only organic customers, the customer arrivals follow a Poisson process. In our base scenario, customer valuations are uniformly distributed within the interval [0, 1], and those arriving with independent needs follow the same distribution. Consequently, the seller maximizes the organic intensity $\lambda^0 = \max_{\lambda\geq 0} \lambda p(\lambda) = \max_{\lambda\geq 0} \lambda(1-\lambda/0.392) = 0.196$ customers per minute. This λ^0 represents the intensity of Poisson arrivals and serves as the lower bound for the optimal intensity. We denote the expected profit under the arrival intensity λ^0 as $V_t^{Myopic}(x)$ and define Profit Loss % $= (V_t(x) - V_t^{Myopic}(x))/V_t(x) \times 100\%$. The following figures display the profit loss associated with neglecting customers' self-excitement.

From Figure 2(a), we observe that ignoring self-excitement can result in a profit loss of up to roughly 13%. Furthermore, the profit loss increases with time-to-go, eventually converging to an upper limit. Furthermore, we note that as the excitement level x increases, the profit loss slightly increases. This is due to the self-exciting function's increasing property. As the excitement level x increases, the gap between the optimal intensity and the organic intensity increases, and thus the profit loss increases.

Figure 2(b) shows that the profit loss of ignoring self-excitement decreases as the decay rate β increases. Specifically, when $\beta = 0.1$, the profit loss is up to 20%, whereas when $\beta = 0.3$, the profit loss falls below 1%. This phenomenon occurs because a higher decay rate implies that customers are more forgetful and quickly cease to excite other customers. Consequently, the word-of-mouth effect diminishes, bringing the customer arrival process close to a Poisson process. Figure 2(c) illustrates that the profit loss increases as α rises. For instance, when $\alpha = 0.25$, the profit loss reaches up to 18%. However, with $\beta = 0.15$, the profit gap remains below 3.2%. This indicates that as the excitation intensity decreases, customers are less likely to excite others, reducing the word-of-mouth effect and thereby decreasing the profit loss of ignoring self-excitement. Consequently, lower excitation intensity results in less profit loss.

Based on the above observations, pricing managers are advised to employ advanced analytics tools to model and calibrate the self-excitement process, thereby helping them understand how current sales influence future demand. Additionally, they are better off ensuring that their pricing policies account for self-exciting behavior, such as the word-of-mouth effect, because ignoring these factors can result in significant profit losses.

6.2. Profit Loss of Static Pricing

Next, we consider an alternative heuristic policy in which the seller sets the price statically based on the expected number of customer arrivals, including organic and self-excited customers. Based on the JD.com data, with 279 customers arriving over 613 minutes, the average arrival rate is $279/613 \approx 0.455$ customers per minute. Assuming customers are uniformly distributed in the interval [0,1] as in our base numerical setting, the optimal static arrival rate is calculated as $\lambda^{Static} = 0.455/2 \approx 0.228$ customers per minute. We denote by $V_t^{Static}(x)$ the expected profit under the arrival intensity λ^{Static} and define Profit Loss % $= (V_t(x) - V_t^{Static}(x))/V_t(x) \times 100\%$. As in the previous section, we can evaluate the performance of the static pricing heuristic.

Figure 2. (Color online) Percentage Profit Loss of Myopic Pricing

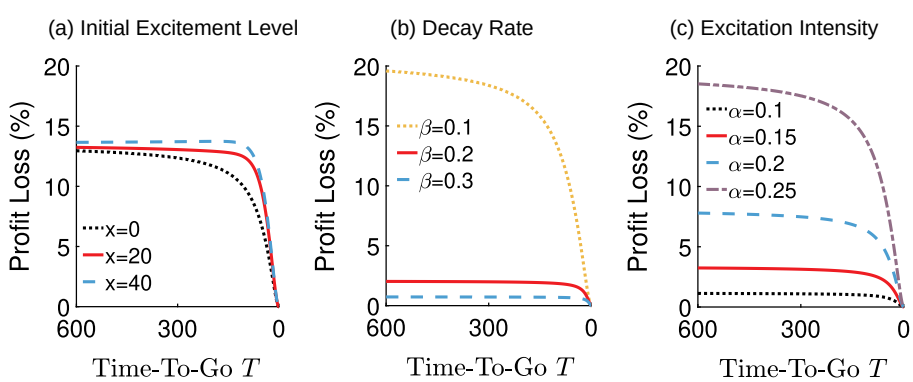


Figure 3. (Color online) Percentage Profit Loss of Static Pricing

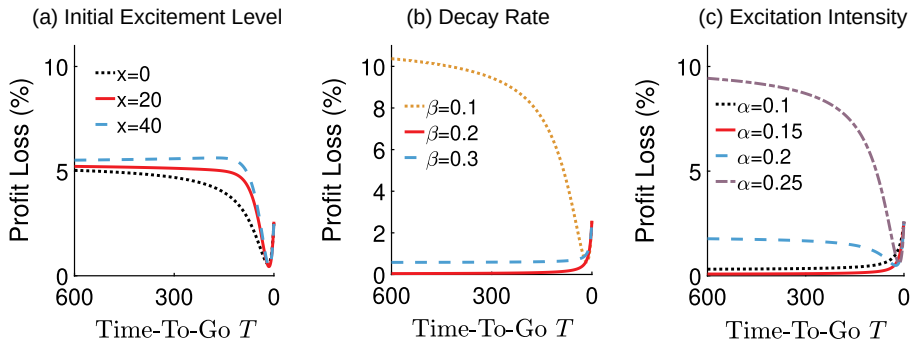


Figure 3(a) indicates that determining the price statically based on the expected number of customer arrivals results in an approximately 5% profit loss. However, as the remaining time decreases, the profit loss initially decreases and then increases. This down-then-up behavior is driven by the observation that the average expected optimal intensity over time is higher as more time remains in the sales horizon for any given initial excitement level. When the time-to-go is long enough, the average expected optimal intensity over the remaining time exceeds the static price, but decreases and approaches the static price as the time-to-go becomes shorter, thereby reducing the profit loss of using the static price. Once the average expected optimal intensity over the remaining time falls below the static intensity and deviates further from it as time runs out, the profit loss begins to increase.

Figure 3(b) illustrates that the profit loss from adopting optimal static pricing does not necessarily follow a monotonic trend as the decay rate β increases. Specifically, when $\beta = 0.2$, the profit loss is below 2%, which is lower than the profit loss observed for both $\beta = 0.1$ and $\beta = 0.3$. This occurs because a higher decay rate indicates that customers quickly lose interest and stop influencing others, thereby reducing the word-of-mouth effect and decreasing the optimal intensity. The average expected optimal intensity for $\beta = 0.2$ is closer to the optimal static intensity compared with $\beta = 0.1$ and $\beta = 0.3$. As a result, the performance for $\beta = 0.2$ surpasses that of both $\beta = 0.1$ and $\beta = 0.3$. Figure 3(c) shows that the profit loss from optimal static pricing does not necessarily exhibit a monotonic trend as the excitation intensity α increases. For example, the profit loss for $\alpha = 0.15$ is lower compared with $\alpha = 0.1$ and $\alpha = 0.2$ when the remaining time is long. This occurs because, as the excitation intensity decreases, customers are less likely to excite others, thereby diminishing the word-of-mouth effect and reducing the average expected optimal intensity over the remaining time. The expected optimal intensity for $\alpha = 0.15$ is closer to the optimal static intensity than $\alpha = 0.1$ and $\alpha = 0.2$, resulting in better performance of adopting optimal static pricing.

As shown above, the static pricing approach can lead to a significant profit loss. Adopting dynamic pricing can help companies maximize profits by adjusting prices in response to real-time customer arrivals and information dissemination via social networks.

6.3. Profit Loss of Deterministic Heuristic

In this section, we adopt the optimal intensity over time from the deterministic problem, referred to as the deterministic heuristic. By solving the associated ordinary differential equations, we find that the deterministic intensity over time depends solely on time and can be computed numerically using the standard forward-backward sweep algorithm. The time interval $[0, T]$ is discretized with a step size δ. In each iteration, we update x_t, λ_t, and μ_t according to the first-order Hamiltonian conditions. The algorithm stops when x_t, λ_t, and μ_t all converge. Details are provided in Algorithm 1.

Algorithm 1 (Forward-Backward Sweep Algorithm for the Deterministic Problem)

Initialize with $\lambda_t = \lambda^*$, $x_t = 0$ and $\mu_t = 0$ for any $t = 0, \delta, 2\delta, \ldots, T$.
Initiate $GAP = 1$
while $GAP \geq 0.001$ **do**
  Calculate $x_{t+\delta}^{new} = x_t^{new} + \alpha n \lambda_t [1 + h(x_t^{new}/n)] - \beta x_t^{new}$ for $t = 0, \delta, 2\delta, \ldots, T - \delta$, with $x_0 = 0$.
  Calculate $\mu_{t-\delta}^{new} = \mu_t^{new} + r(\lambda_t) h'(x_t^{new}/n) + [\alpha \lambda h'(x_t^{new}/n) - \beta] \mu_t^{new}$ for $t = T, T - \delta, \ldots \delta$, with $\mu_T^{new} = 0$.
  Calculate λ_t^{new} such that $r'(\lambda_t^{new}) + \alpha \mu_t^{new} = 0$ for $t = 0, \delta, 2\delta, \ldots, T$.
  Update $GAP = \max\{\sum_t |\lambda_t - \lambda_t^{new}| / \sum_t \lambda_t^{new}, \sum_t |x_t - x_t^{new}| / \sum_t x_t^{new}, \sum_t |\mu_t - \mu_t^{new}| / \sum_t \mu_t^{new}\}$.
  Update $\lambda_t = \lambda_t^{new}$, $x_t = x_t^{new}$ and $\mu_t = \mu_t^{new}$.
end while

We denote by $V_t^D(x)$ the expected profit under the arrival intensity λ^D and define Profit Loss % $= (V_t(x) - V_t^D(x))/V_t(x) \times 100\%$. As in the previous section, we can evaluate the performance of the deterministic pricing heuristic.

Figure 4. (Color online) Percentage Profit Loss of Deterministic Heuristic

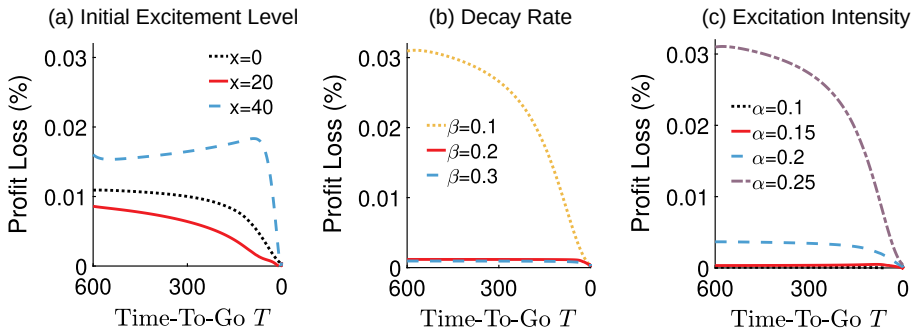


Figure 4(a) demonstrates that the time-varying deterministic heuristic significantly outperforms both myopic and optimal static pricing heuristics because it can, to some degree, capture the time-varying nature of the optimal price trajectories (Theorem 3), whereas both myopic and static pricing cannot. The maximum profit loss is less than 0.02%, 0.012%, and 0.009% for initial excitement levels $x = 0$, $x = 20$, and $x = 40$, respectively. Figure 4(b) shows that the profit loss of the deterministic heuristic decreases as the decay rate increases. A higher decay rate means that customers quickly lose interest and stop influencing others, diminishing the word-of-mouth effect and shifting the customer arrival process closer to a Poisson process, which has a lower variability than a general Hawkes process driven by self-excitement. Consequently, the deterministic heuristic is closer to the (random) optimal intensity of the stochastic problem. Figure 4(c) indicates that the profit loss of the deterministic heuristic increases as the excitation intensity increases. As the excitation intensity decreases, earlier buyers are less likely to entice additional arrivals, weakening word-of-mouth effects and causing the arrival process to approach a Poisson process. For the same reason mentioned above, a lower excitation intensity results in less profit loss when adopting deterministic heuristics.

Our numerical experiments confirm the deterministic heuristic's superior performance, which is theoretically demonstrated in Theorem 4. The deterministic heuristic simplifies the pricing decision-making process, allowing pricing managers to make well-performed pricing decisions efficiently.

7. Conclusion

In this paper, we investigate how a seller can dynamically adjust selling prices under a generalized self-exciting customer arrivals model, where each purchase stimulates additional purchases, increasing the arrival intensity. Self-excitement complicates the optimal pricing policy. We identify the excitement level as the key sufficient statistic and fully characterize the structural properties of the optimal policy. Our findings indicate that the optimal pricing policy encourages higher purchase rates as excitement levels rise and that the optimal price trajectory exhibits an upward trend in the absence of excitation decay. Moreover, we examine the deterministic version of our problem. We show that the performance loss of the deterministic time-varying (open-loop) heuristic compared with the optimal profit is just a constant. We also connect the immigration-birth process and extend our base model to account for discrete pricing and multiple products, offering comprehensive insights into dynamic pricing strategies under various market conditions. Finally, our numerical results highlight the critical importance of accounting for self-excitement in pricing decisions, as neglecting this factor leads to significant profit loss.

To our knowledge, this is the first study to model customer arrivals as a self-exciting process in RM, capturing how past purchases influence future customer arrivals. Our findings reveal that the optimal price trajectory tends to increase over time, in contrast to the existing RM literature, where the optimal price tends to remain constant without capacity constraints. Perhaps surprisingly, we find that the optimal price's reaction to the excitement level depends on whether the market is in a growth or saturation phase. Additionally, we adopt the deterministic heuristic to the stochastic problem and demonstrate that its performance loss is a constant, in contrast to traditional RM problems.

We point out some directions for future research. The first direction involves introducing the capacity constraint, a feature of the classic RM problems. The second direction is to consider separate streams of organic customers, each for one product in a category, but a purchase of one product may excite interest in other products.

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Endnotes

[1] Caro and Martínez-de Albéniz (2020) similarly model the decay in the attractiveness of content released over time.

[2] The intensity $\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$ is originally introduced by Hawkes (1971). They use the exponential decay rate to describe the self-exciting process. The generalized self-exciting intensity is introduced by Brémaud and Massoulié (1996) and Zhu (2015). The bivariate marked self-exciting process of interaction between customers and agents is modeled by Daw et al. (2025).

[3] These monotonicity results do not hold under local convexity or concavity. A counterexample is provided in Online Appendix A.1.

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